



COA 2.0

MAIN MEMORY: RAM & ROM

By: Vishvadeep Gothi



→ Memory

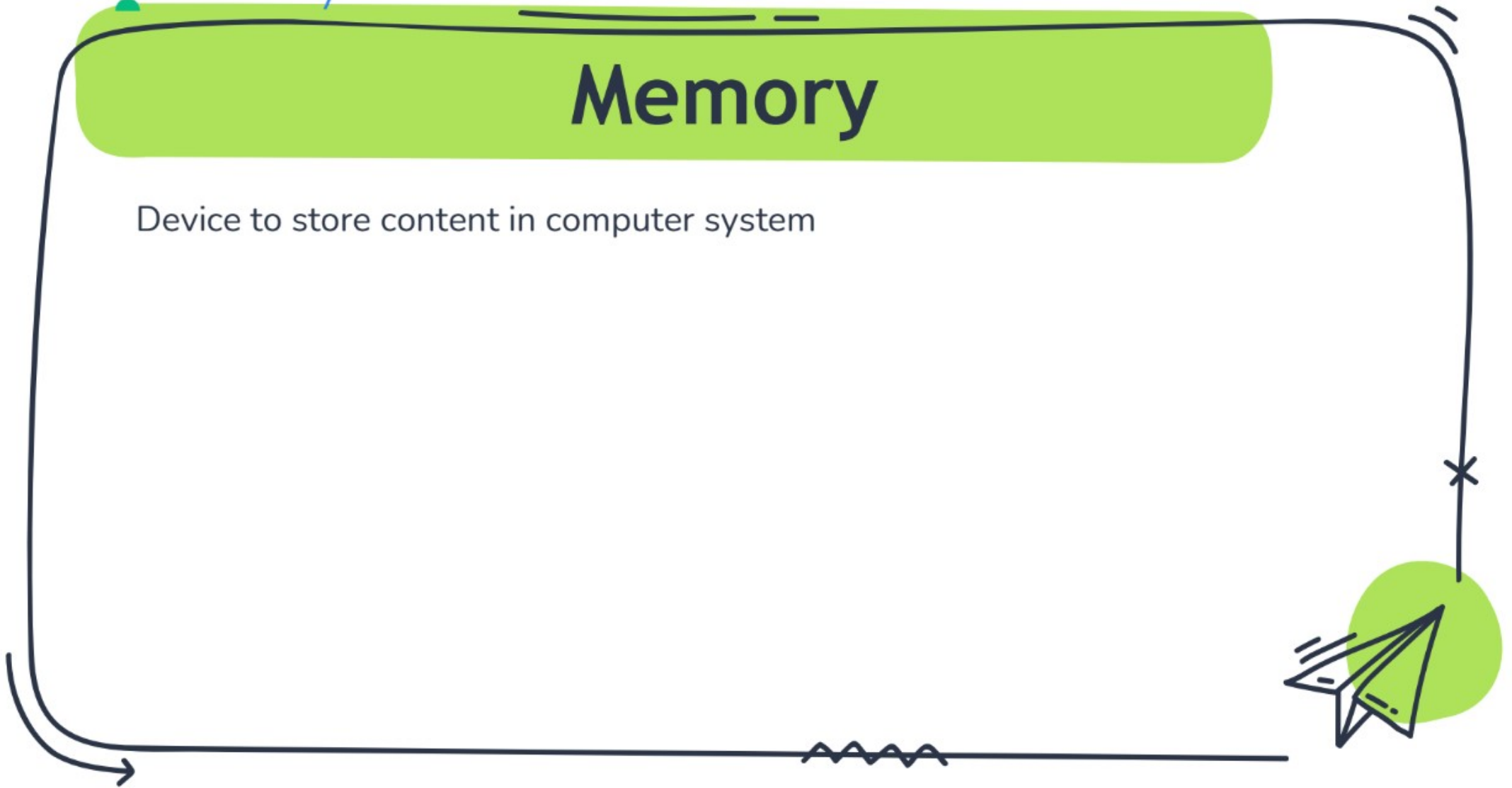
→ Representatⁿ: no. of cells $\times$ 1 cell capacity

Are You Ready?

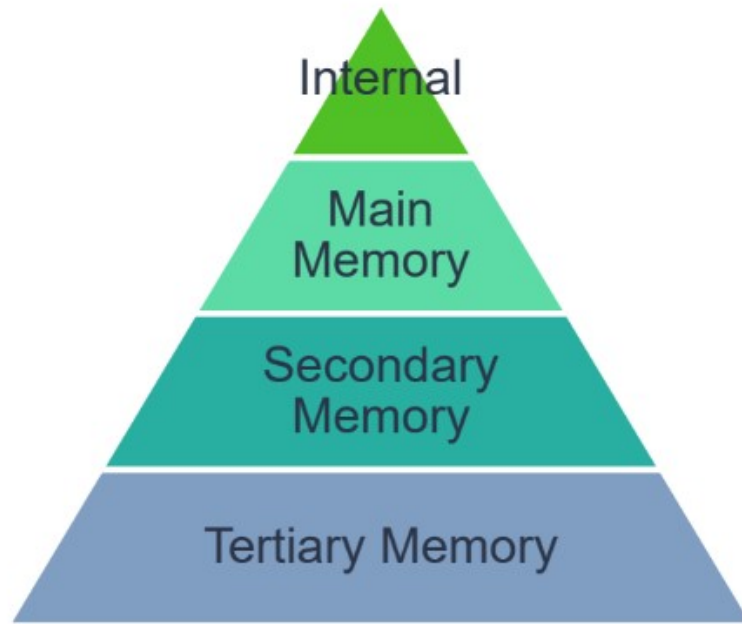
Let's Begin the Unlimited Learning

Memory

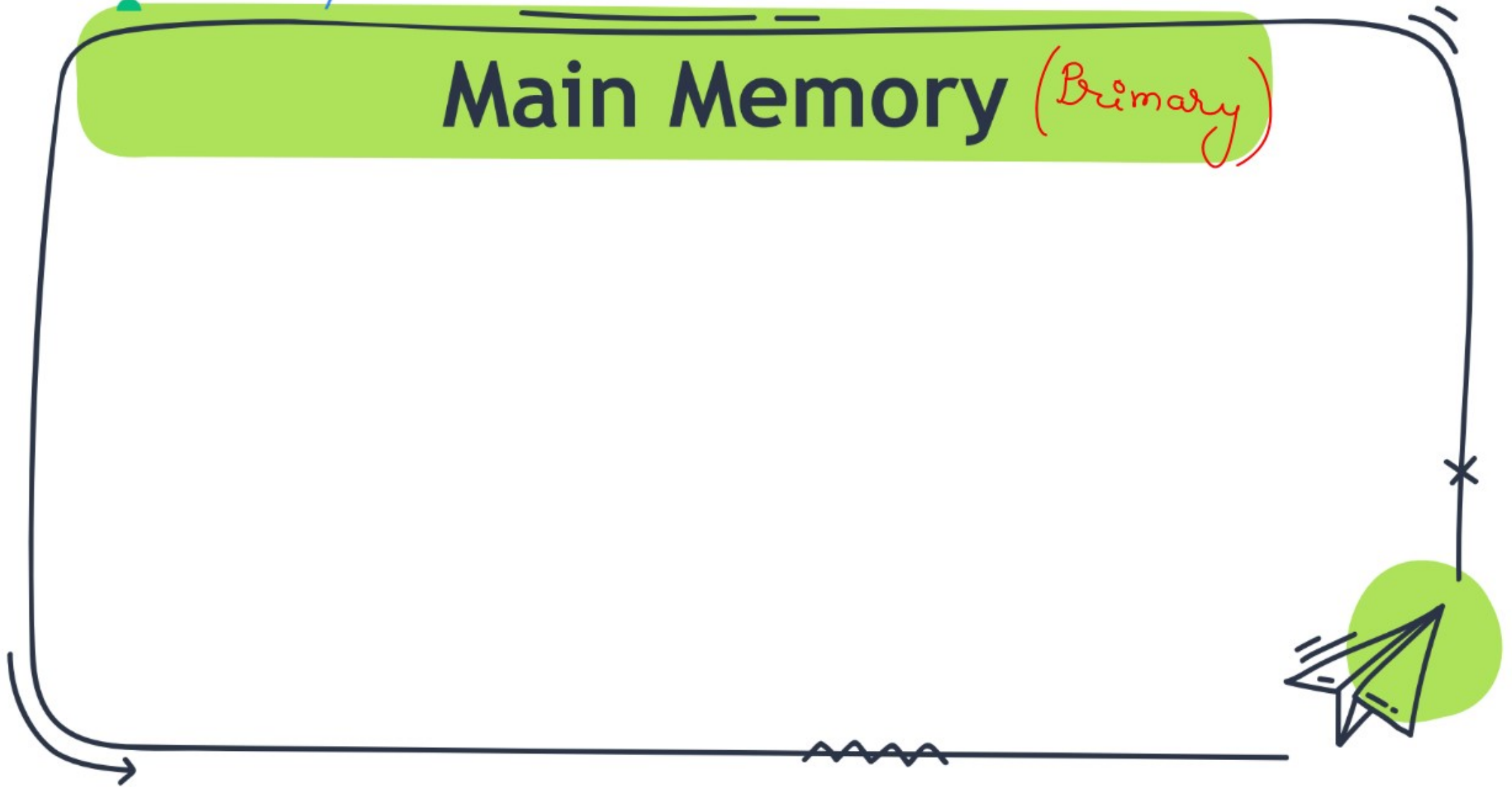
Device to store content in computer system



Memory Hierarchy



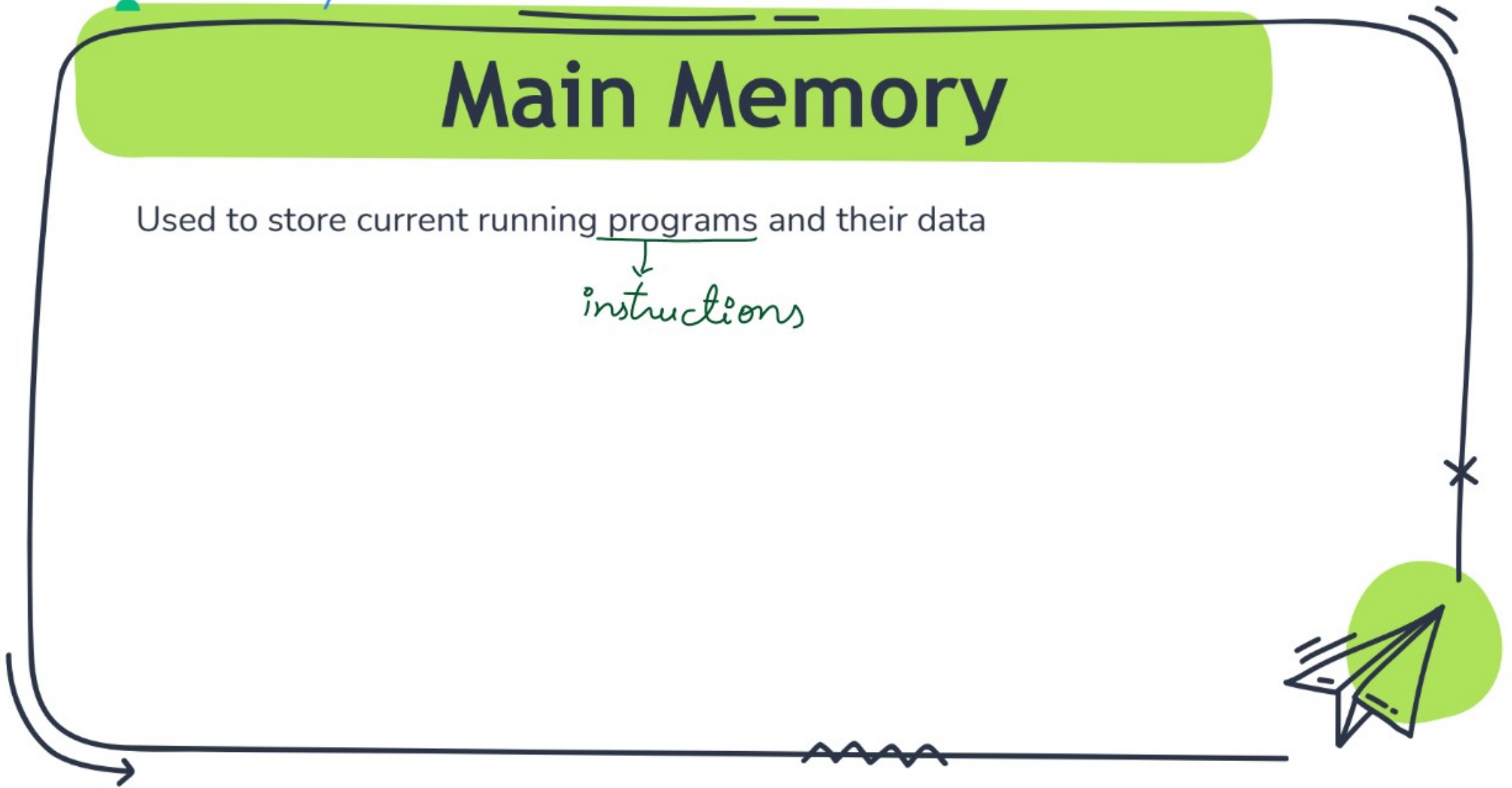
Main Memory (Primary)



Main Memory

Used to store current running programs and their data

↓
instructions



Main Memory

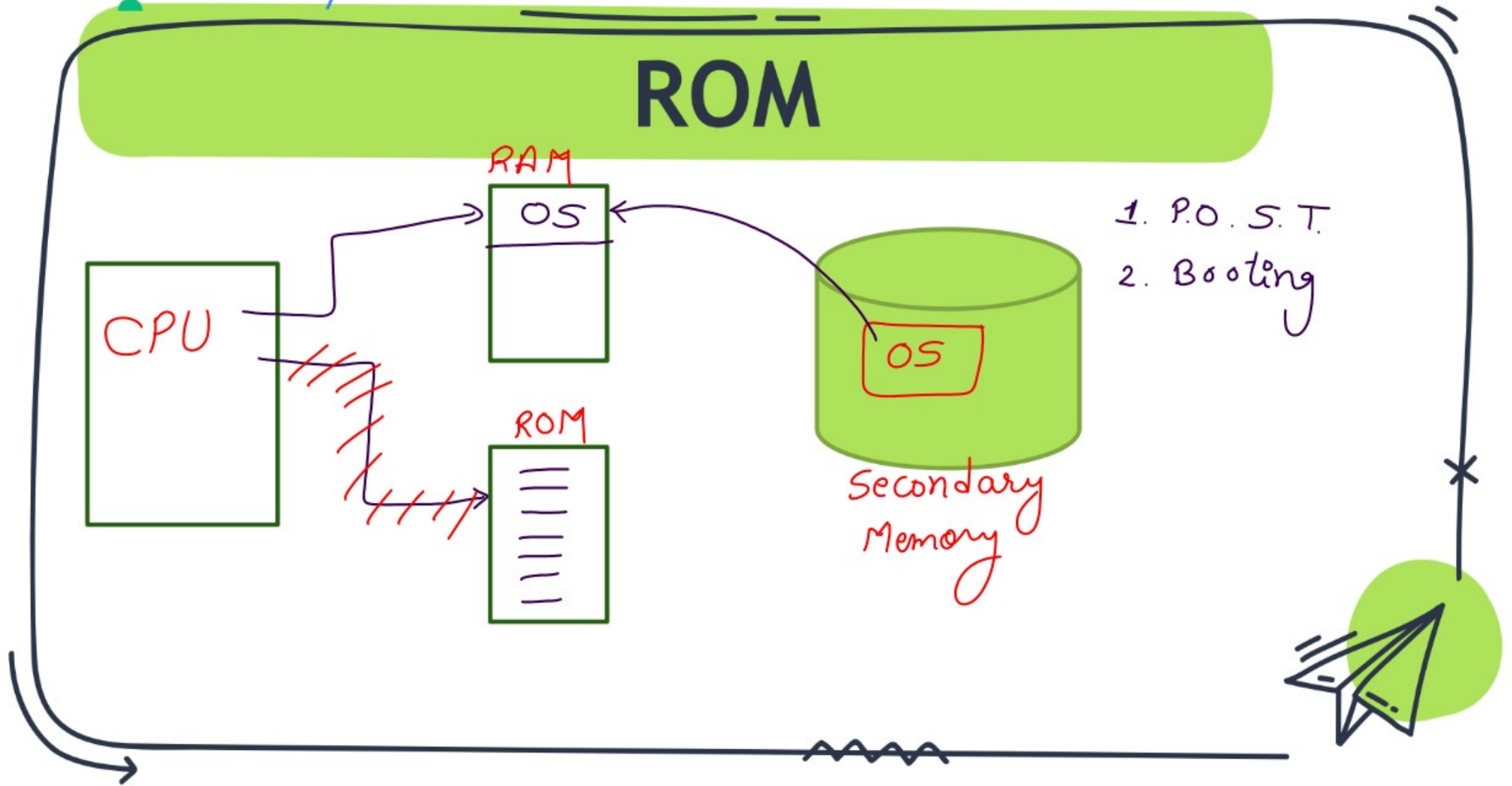
Used to store current running programs and their data

Types:

1. RAM (Random Access Memory) $\Rightarrow$ volatile
2. ROM (Read only Memory) $\Rightarrow$ non-volatile



ROM

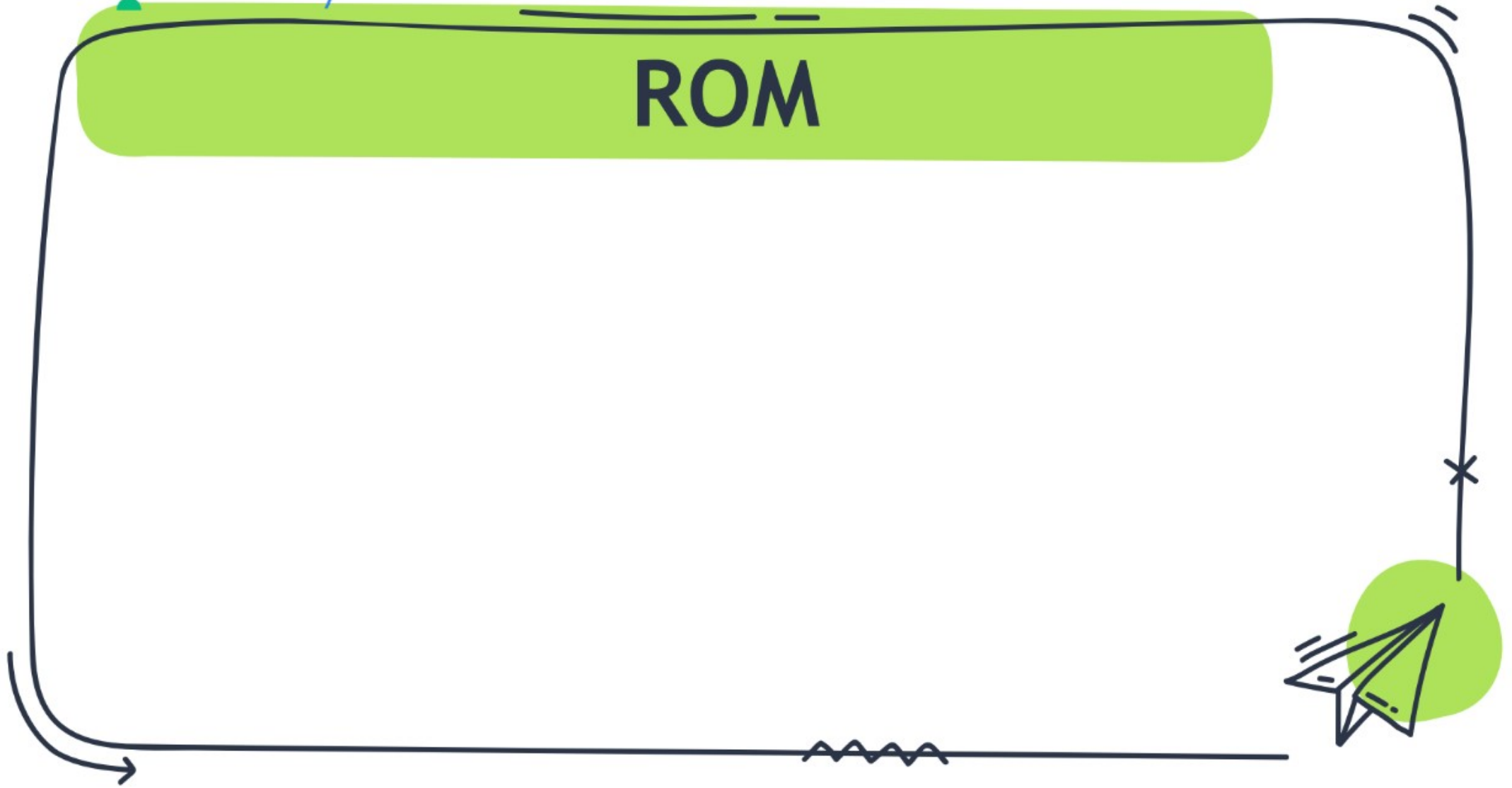


ROM

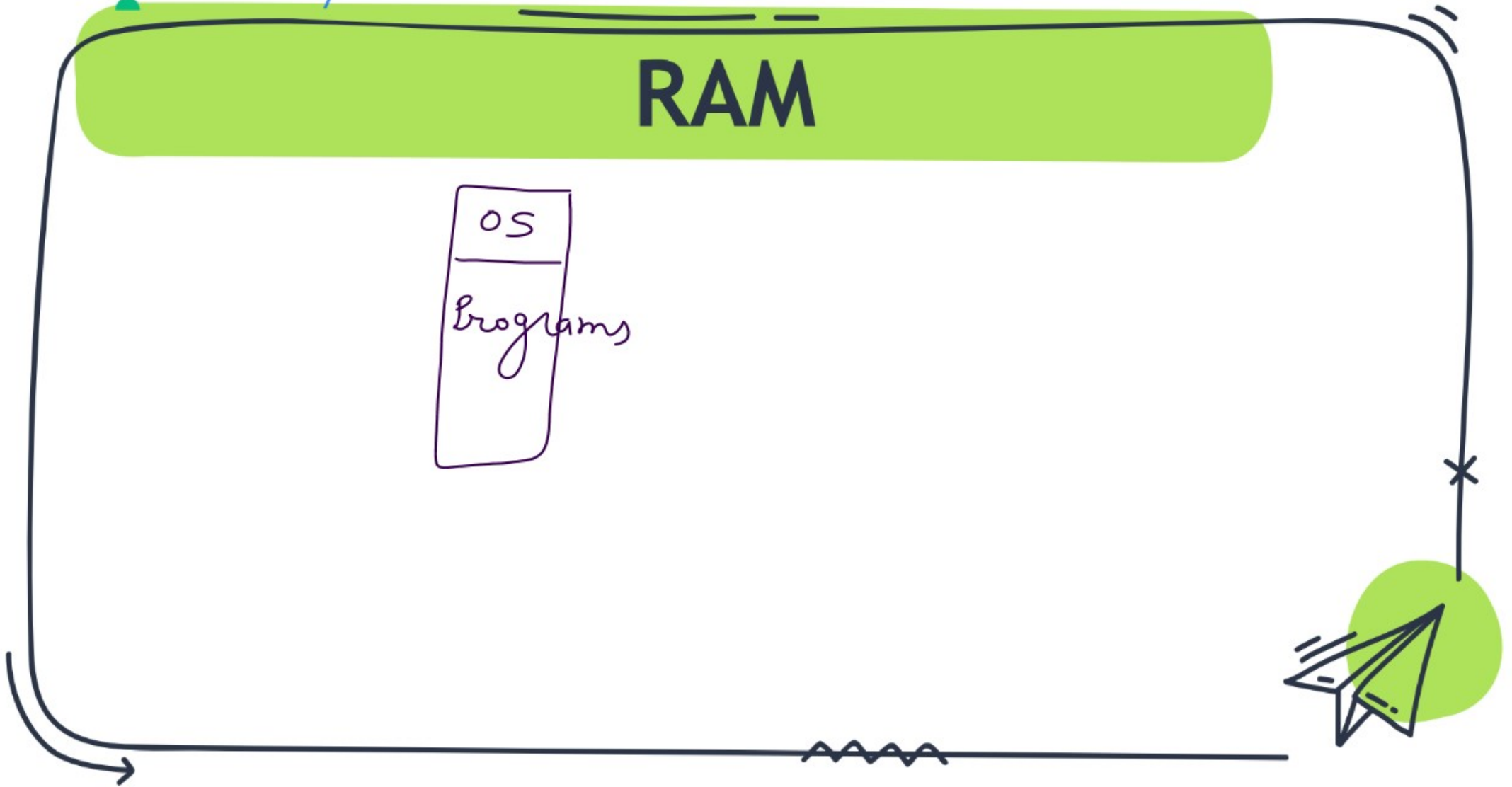
A program, execution of which is responsible for booting is known as *Bootstrap program*.



ROM



RAM



Types of RAM

Static (SRAM)	Dynamic (DRAM)
1. Made of $\rightarrow$ flip-flops 2. No refresh 3. Faster 4. Expensive 5. Used for cache	1. Capacitors $\rightarrow$ storage in form of electric charges. 2. Periodic Refresh is required to keep content. 3. slower 4. Less Expensive 5. Used for m.m. implementation



Types of RAM

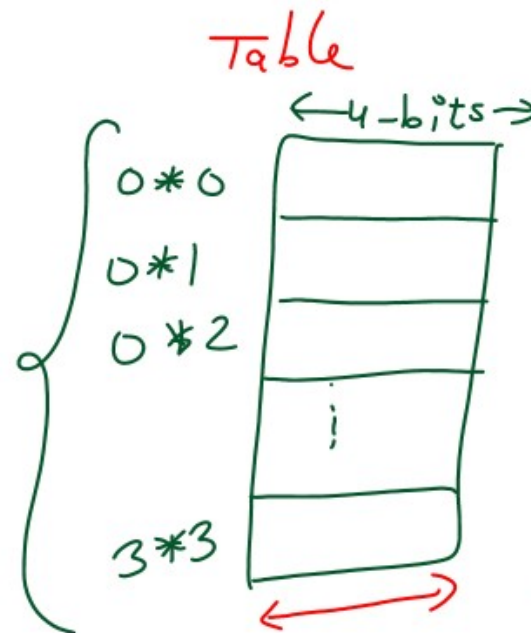
Static	Dynamic
1. Implemented using flip-flops	1. Implemented using capacitors
2. No refresh required	2. Periodic refresh is required
3. Faster Read/Write	3. Slow Read/Write
4. Used for Cache	4. Used for main memory
5. Low Idle power consumption	5. High Idle power consumption
6. High operational power consumption	6. Low operational power consumption



Table for Multiplication

Table of multiplication of two 2-bits unsigned integers.

<u>a</u> 2-bits	<u>b</u> 2-bits
00	00
01	01
10	10
11	11



Total no. of
outputs
to be
stored $= 2^2 * 2^2$
 $= 16$

Table $\Rightarrow 16 \times 4\text{-bits}$
 $2^n \times 2n\text{-bits}$



Table for Multiplication

Input 1	Input 2	Multiplication Result Size	Multiplication Table Size	Addition Result	Addition Table Size
4-bits	4-bits	8-bits	$2^8 \times 8\text{-bits}$	5-bits	$2^8 \times 5\text{-bits}$
$n\text{-bits}$	$n\text{-bits}$	$2n\text{-bits}$	$2^{2n} \times 2n\text{-bits}$	$(n+1)\text{bits}$	$2^{2n} \times (n+1)\text{-bits}$

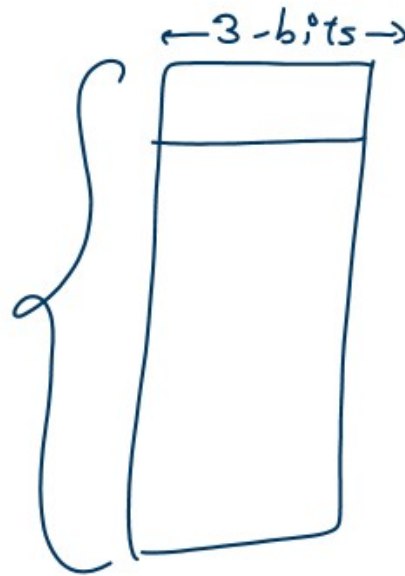


Table for ~~Multiplication~~

Addition

$\frac{a}{2\text{-bits}}$	$\frac{b}{2\text{-bits}}$
00	00
01	01
10	10
11	11

2^4



$2^4 \times 3\text{-bits}$

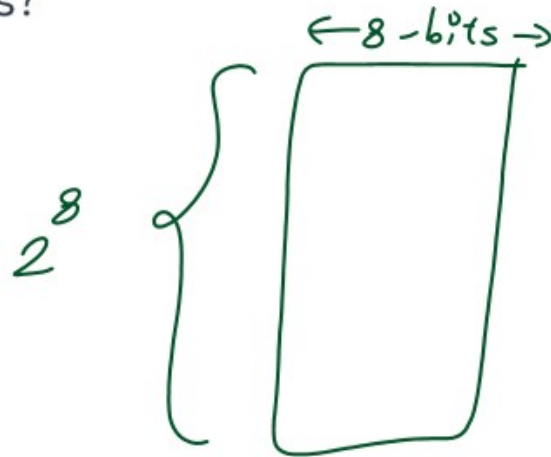
$2^n \times (n+1)\text{ bits}$



Question 1

The amount of ROM needed to store the table for multiplication of two 4-bit unsigned integer is?

- a) 64 bits
- b) 128 bits
- c) 1K bits
- d) 2K bits



$2^8 \times 8\text{-bits}$

2^{11} bits

2K bits



Question 2

A ROM is used to store the table for multiplication of two 8-bit unsigned integers. The size of ROM required is?

- a) 256×16
- b) $64K \times 8$
- c) $4K \times 16$
- d) $64K \times 16$

$$2^{16} \times 16 \text{ bits}$$
$$64K \times 16 \text{ bits}$$

